

Michael Black

Engineering Software Developer / Business Analyst

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Civil engineer focused on improving how teams use engineering software through better workflows, tools, and guidance.

Profile

Civil engineer with 15+ years of experience across design, BIM coordination, and digital operations, with a focus on engineering software, workflows, and user adoption.

Consistently takes ownership of new tools—evaluating what they can and cannot do, defining how they should be used, and helping teams apply them effectively in real project environments.

Experience includes developing internal software tools, implementing BIM and VR workflows, and supporting large design teams through documentation, guidance, and hands-on training.

Core Strengths

- **Engineering Software:** Civil 3D, Navisworks, BIM workflows, ProjectWise environments
- **Workflows & Processes:** Translating technical systems into clear, repeatable processes
- **Training & Support:** User onboarding, live training, troubleshooting, team support
- **Documentation:** Step-by-step guides, workflow standards, user-facing instructions
- **Technology Evaluation:** Assessing tools, defining capabilities, identifying limitations
- **Automation:** Python, Dynamo, LISP for workflow and production improvements

Professional Experience

Analyst, Digital Operations – WSP (2023 – Present)

- Contributed to the development of the Digital Operations Center (DOC), an in-house C# / Windows Forms application supporting large project teams
- Centralized access to required software (Civil 3D, OpenRoads, etc.) and ensured correct versions were installed and maintained
- Reduced risk of data corruption caused by non-compliant software environments

- Supported development of Civil 3D client environments, including templates, profiles, and standards
- Developed automation tools using Dynamo and LISP to improve production workflows
- Provided Navisworks clash detection support, including setup guidance and coordination support
- Tested and evaluated Autodesk InfraWorks, Bentley iTwin, and Revizto for potential project use
- Produced summary documentation outlining capabilities, limitations, and recommended use cases for each platform
- Tested and configured model coordination platforms (iTwin, BIMTrack), including workflow evaluation and documentation
- Produced user guidance and support material to help teams understand and apply tools and workflows

Civil 3D Specialist – Union Station Enhancement Project, WSP (2021 – 2023)

- Supported design teams in resolving technical issues in drawings and models, including Civil 3D and ProjectWise integration challenges
- Played a lead role in implementing Virtual Reality (VR) workflows for design model reviews
- Defined model preparation requirements and created supporting documentation and guidance
- Developed Python scripts for automated DWG quality checks
- Conducted BIM integration testing across platforms (InfraWorks, iTwin), including documenting workflows and findings
- Provided ongoing technical support to improve model quality and team efficiency
- Rolled out VR design reviews on an Alliance contract model project and Union Station Enhancement Project, including setup of corporate VR room in the Toronto office
- Created comprehensive VR guidance documents: “How to use VR, what to watch out for, safety protocols,” and “How to prepare design models from various software for VR review”

BIM Coordinator / VR Specialist – SNC-Lavalin (2019 – 2021)

- Acted as BIM Team Lead for Navisworks clash detection and model coordination across rail infrastructure projects
- Set up models, clash tests, and search sets across multiple project segments and stations

- Led weekly coordination meetings and reported design status to senior management
- Led implementation of Virtual Reality (VR) model reviews across the project
- Defined VR workflows and hosted VR sessions with multidisciplinary teams
- Worked with software developers (IrisVR) to test features and provide user feedback
- Created training materials and conducted sessions on VR software usage and workflows
- Authored the company-wide Navisworks Clash Detection Guide
- Developed corporate VR procedure, including setting up a VR room in the Toronto office and defining standardized VR review protocols
- Produced VR learning tools documenting preparation of design models from multiple software platforms for VR use, and outlining safety, usage, and review best practices

Intermediate Land Development Engineer – IBI Group (2012 – 2019)

- Designed subdivision infrastructure including grading, storm/sanitary networks, and utility coordination
- Transitioned workflows from manual processes to Civil 3D-based automation
- Implemented Civil 3D Quantity Takeoff workflows to replace manual calculations
- Became a key resource for Civil 3D support, assisting team members with tools and workflows
- Improved team efficiency through better use of engineering software and processes

Education

Bachelor of Engineering (Civil Engineering) – Toronto Metropolitan University (Ryerson University), 2009 Tools & Technology

- **Engineering Software:** Civil 3D, Navisworks, ProjectWise, BIMTrack, InfraWorks, iTwin
- **VR:** Prospect (IrisVR)
- **Programming:** Python, Dynamo, LISP, HTML/JavaScript